

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1511

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3) Total Marks: 40
Question Count: 4

Code of Conduct

I **V.PREETE (192572004)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V.PREETE

Experiment 1:

Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

ANSWER:

Library book reservation system

Trial expires in 11 days [Upgrade](#) [Edit this application](#) [Help](#)

Library book reservation system

STUDENT ID registration number

112833	19256890
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student name appu

mobile number +912233557799

book name python programming

author john zella

category computer science

shelf number 3

status available

[Add a comment](#)

Experiment 2:

Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

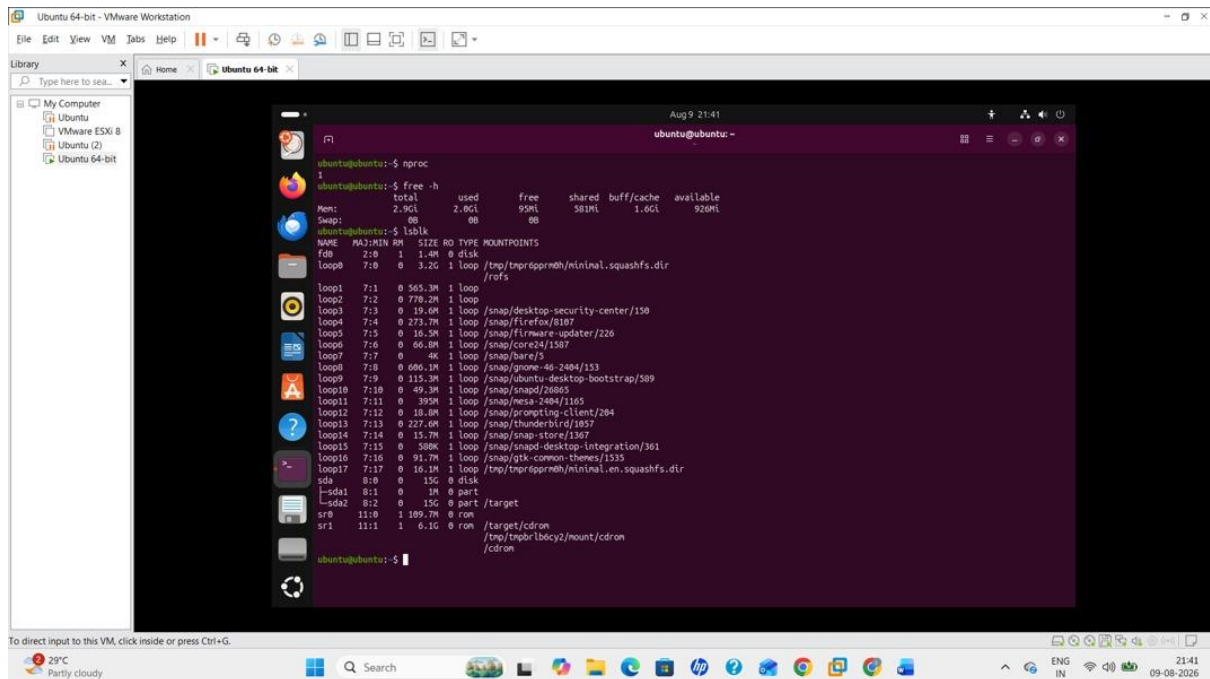
ANSWER:

The screenshot displays a web application for a Payroll Processing System. On the left is a dark sidebar with navigation options: 'Payroll Processing System' (selected), 'Payroll Processing System', and 'Payroll Processing System Report'. The main content area is titled 'Payroll Processing System Report' and features a table with two columns: 'Employee name' and 'Payment Status'. The first row shows 'Employee ID' 112345. Below the table, there is a section for 'Add a comment' and a status 'Showing 1 of 1'. The bottom of the screen shows a user profile for 'PREETEV'.

Employee name	Payment Status
Employee ID	112345
Department	CSE
Designation	App developer
Overall Experience	10
Basic Pay	₹ 30,000.00
Net Pay	₹ 4,000.00
Gross Pay	₹ 42,000.00
Professional Tax	₹ 2,000.00
PF Deduction	₹ 1,800.00
Payment Status	Paid

3. Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

ANSWER:



The screenshot shows a VMware Workstation interface with an Ubuntu 64-bit virtual machine. The terminal window displays the following commands and their outputs:

```
ubuntu@ubuntu:~$ nproc
1

ubuntu@ubuntu:~$ free -h
             total        used        free      shared  buff/cache   available
Mem:          2.0Gi          2.0Gi          0B          0B          0B          0B
Swap:          0B           0B           0B          0B          0B          0B

ubuntu@ubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
fd0          2:0    1    1.4K 0 disk 
loop0        7:0    0    3.2G 1 loop /tmp/tmprppmndh/minimal.squashfs.dir
             /rofs
loop1        7:1    0    565.3M 1 loop 
loop2        7:2    0    778.2M 1 loop 
loop3        7:3    0    19.6M 1 loop /snap/desktop-security-center/150
loop4        7:4    0    273.7M 1 loop /snap/firefox/8187
loop5        7:5    0    16.5M 1 loop /snap/firmware-updater/226
loop6        7:6    0    66.8M 1 loop /snap/core24/1587
loop7        7:7    0    4K 1 loop /snap/bare/5
loop8        7:8    0    666.1M 1 loop /snap/gnome-46-2404/153
loop9        7:9    0    115.3M 1 loop /snap/ubuntu-desktop-bootstrap/589
loop10       7:10   0    49.3M 1 loop /snap/snapd/26865
loop11       7:11   0    395M 1 loop /snap/mesa-2404/1165
loop12       7:12   0    18.8M 1 loop /snap/prompts-client/284
loop13       7:13   0    227.6M 1 loop /snap/thunderbird/1857
loop14       7:14   0    15.7M 1 loop /snap/snap-store/1367
loop15       7:15   0    308K 1 loop /snap/snap-desktop-integration/361
loop16       7:16   0    91.7M 1 loop /snap/gtk-common-themes/1535
loop17       7:17   0    16.1M 1 loop /tmp/tmprppmndh/minimal.en.squashfs.dir
sda          8:0    0    15G 0 disk 
└─sda1       8:1    0    1K 0 part 
└─sda2       8:2    0    15G 0 part /target
sr0          1:0    0    109.7M 0 rom 
sr1          1:1    0    6.1G 0 rom /target/cdrom
              /tmp/tmprlbcy2/mount/cdrom
              /cdrom

ubuntu@ubuntu:~$
```

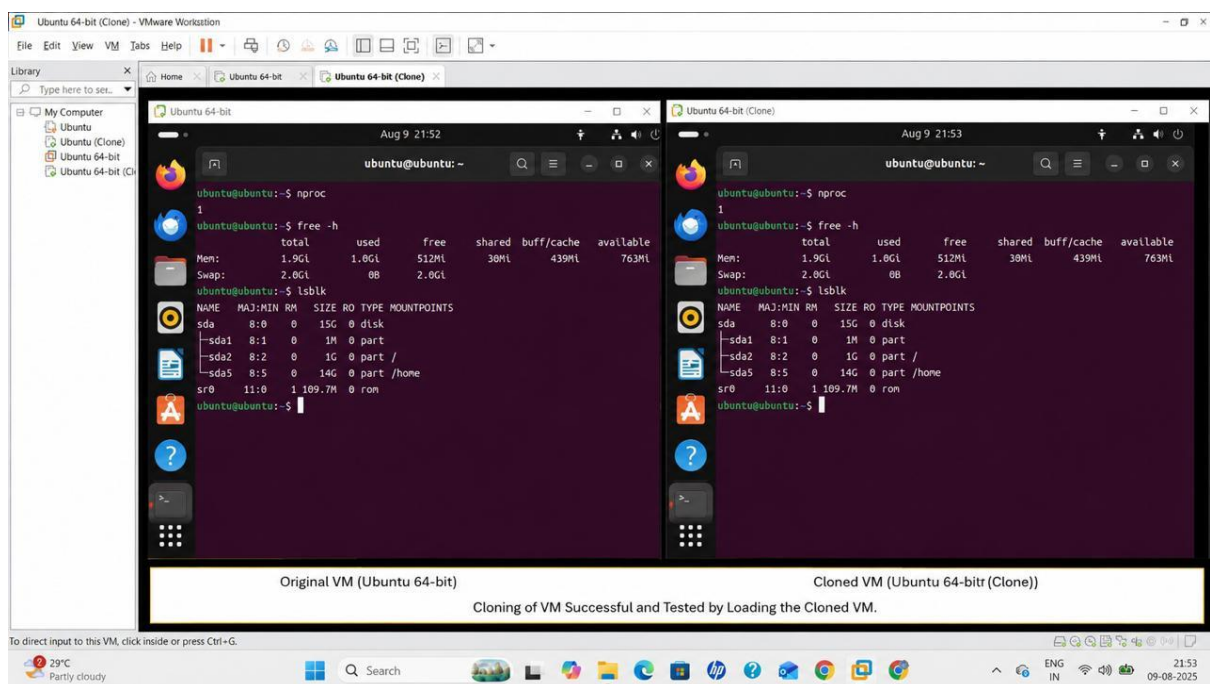
Experiment 4:

Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system

(Either Windows/Linux). Create a Cloning of a VM and Test it by loading the

Previous Version/Cloned VM

ANSWER:



Assessment Tasks

Experiment 1:

Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

ANSWER:

patient name	Prema
patient id	678965
age	65
gender	female
address	no.22 jayaraj nagar, chennai-37
Phone number	+911122334455
Email	prema@gmail.com
attender name	vinu
attender phone no	+918838408006
Disease / Diagnosis	Fever and Infection
blood group	O+
Admission Date	22-Jul-2026 00:00:00
Room / Ward No.	Ward 101
Treatment Details	Medication and observation

Experiment 2:

Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

ANSWER:

The screenshot displays a web application titled "Online Super Market". The interface is divided into three main sections:

- Sidebar:** Contains the application name "Online Super Market" and a link to "Online Super Market Report".
- Main Report Area:** Titled "Online Super Market Report", it shows a list of records. One record is visible for "vinu" with the address "no.14 jayaraj nagar, mogappair, chennai-37".
- Form View:** A detailed form for the selected record "vinu". The form fields and their values are as follows:

Field	Value
customer name	vinu
address	no.14 jayaraj nagar, mogappair, chennai-37
Phone number	+919988776655
Email	vinu@gmail.com
item name	rice
item category	
quantity	5
Unit price	₹ 200.00
payment method	UPI

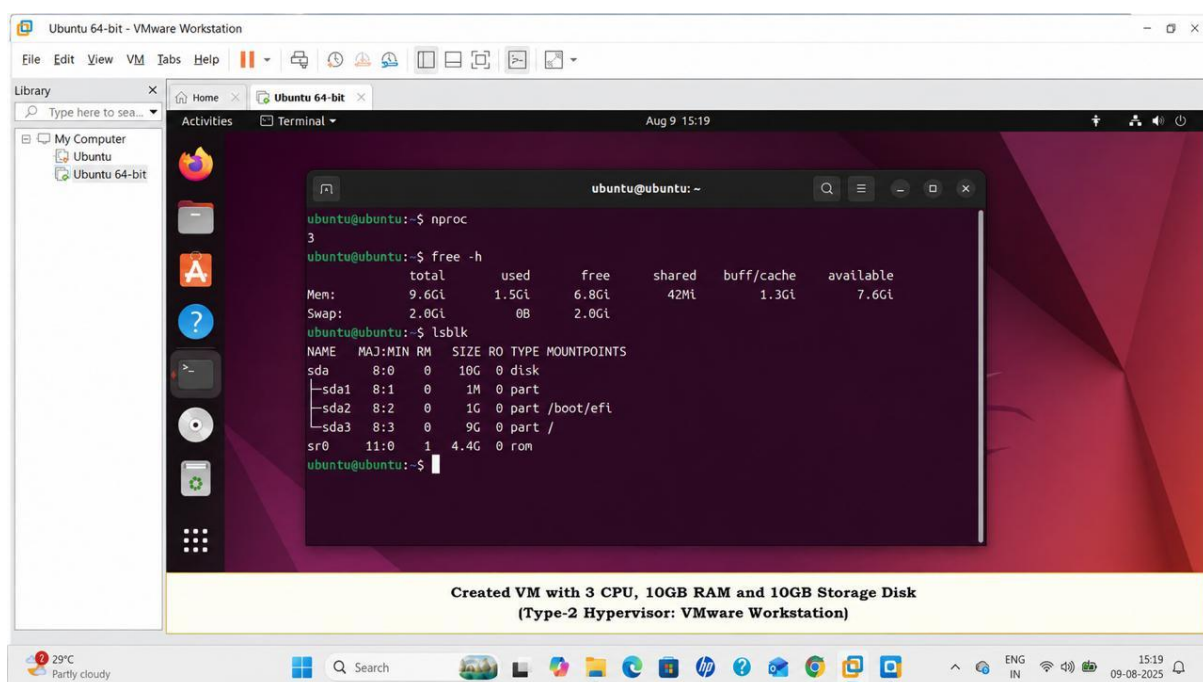
Below the form, there is a link to "Add a comment". The bottom of the interface shows the user's name "PREETE V" and a notification icon.

Experiment 3:

Demonstrate virtualization by Installing Type-2

Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

ANSWER:



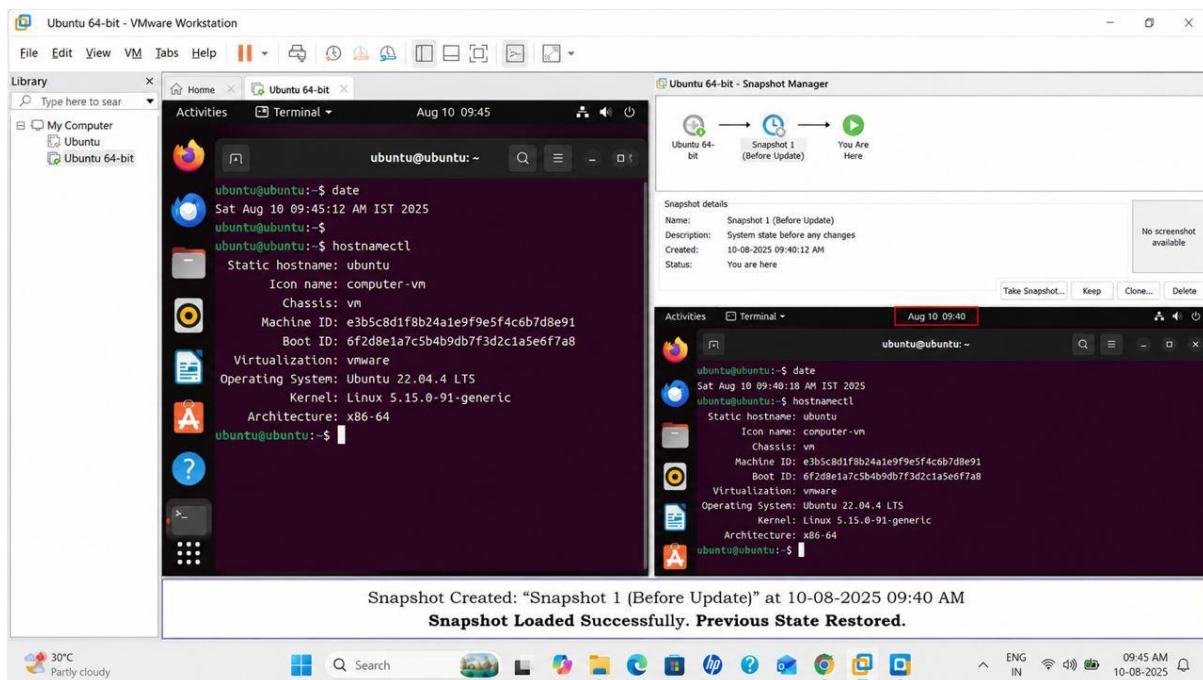
Experiment 4:

Demonstrate virtualization by Installing Type-2

Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot

of a VM and Test it by loading the Previous Version/Cloned VM.

ANSWER:

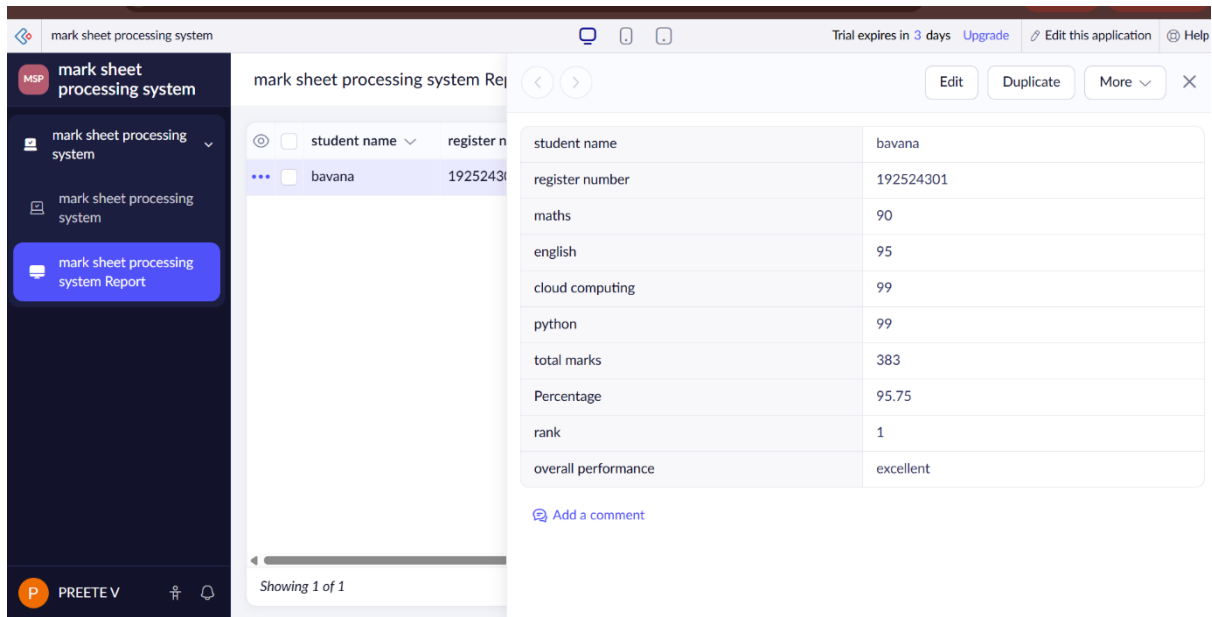


Assessment Tasks

Experiment 1:

Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

ANSWER:



student name	bavana
register number	192524301
maths	90
english	95
cloud computing	99
python	99
total marks	383
Percentage	95.75
rank	1
overall performance	excellent

Experiment 2:

Create a simple cloud software application for

Property Buying & Rental process (In Chennai city) using any

Cloud Service Provider to demonstrate SaaS include the necessary

fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

ANSWER:

The screenshot displays a web application interface for 'Property Buying' in a 'Development' environment. The main content area shows a 'property buying Report' form. The form includes a table with the following data:

customer id	full name
112833	madhu

Below the table, there is a form with the following fields:

- customer id: 112833
- full name: madhu
- gender: Choice 1
- mobile number: +919445123325
- Email: madhu.bavana@gmail.com
- address: no.9jayaraj nagar,anna nagar,chennai-58
- aadhar number: 3456-6789-7365

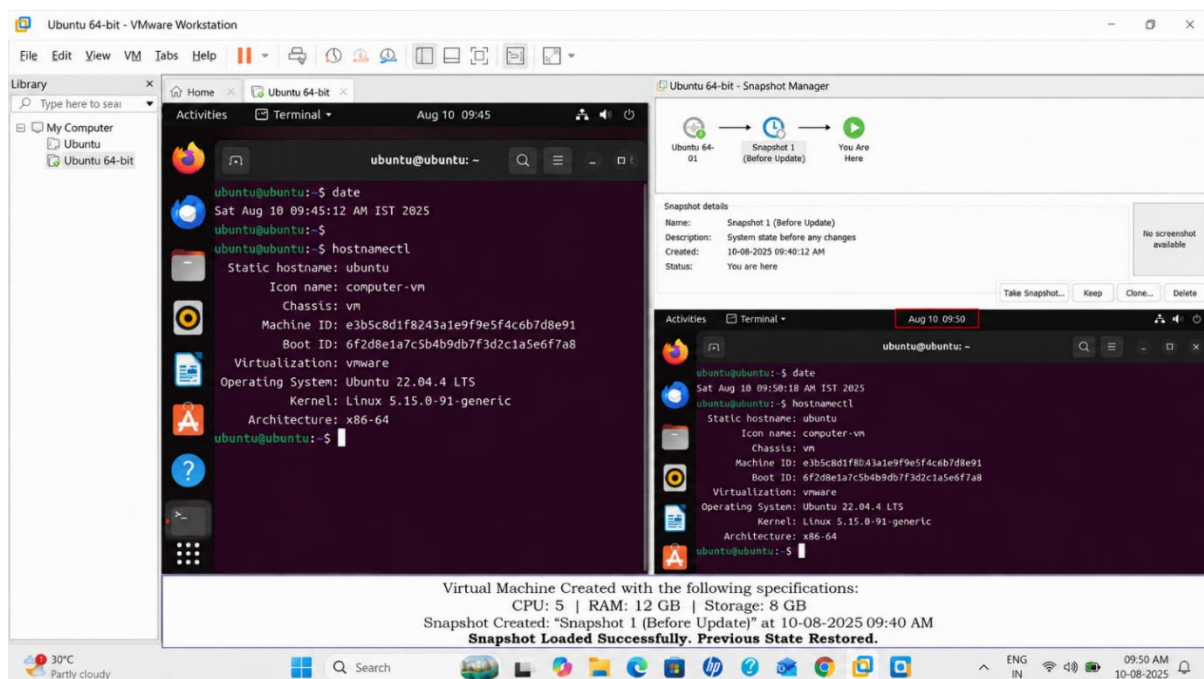
At the bottom of the form, there is a link to 'Add a comment'.

Experiment 3:

Demonstrate virtualization by Installing Type-2

Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

ANSWER:



Experiment 4:

Demonstrate virtualization by Installing Type-2

Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot

of a VM and Test it by loading the Previous Version/Cloned VM.

ANSWER:

